

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – Scenario Based
Questions

Course Code:	DSA0405	Course Title:	FUNDAMENTALS OF DATA SCIENCE
CO Assessed:	CO4-Scenario based questions	Assessment:	Assessment Tool 1– scenario based questions
Weightage:	40% of CIA	Total Marks:	100
CO3 Statement:	Analyze and extract image features using detection and matching techniques for effective image representation and comparison. (BL4)		
Student Name:	M.SREEVARDHAN	Reg. No.:	192472405
Section / Batch:	CSE-AI/2024	Date:	20/08/26

Code of Conduct

I M.SREEVARDHAN/192472405 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____ VAYU _____

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning	Marks
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly	
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts	
Analysis	20	In-depth analysis with strong reasoning	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning	

		and insights				
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided	
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand	

Statistical Hypothesis Testing Assessment

This document outlines the professional analyses and interpretations for the five hypothesis testing scenarios provided.

Assessment #	Scenario Focus		Statistical Outcome	Core Decision
1 (Q16)	Software	Response Time	$p = 0.004 < 0.01$	Support company claim
2 (Q17)	Ad Campaign	CTR	$p = 0.12 > 0.05$	Fail to support increase
3 (Q18)	ML Accuracy	Model	$p = 0.01 < 0.05$	Adopt new model
4 (Q19)	Factory Defect Rate	Defect	CI: (2.4%, 3.6%), $p = 0.01$	Reject 2% claim
5 (Q20)	UI Design	Revenue	CI: (₹120, ₹480), $p = 0.002$	Deploy new UI

Detailed Analyses:

Analysis 1: Software Application Response Time

Scenario Context: A software company claims its new application has an average response time of less than 2 seconds. The hypothesis test yields a p-value of 0.004, with a threshold $\alpha = 0.01$.

80/20 Core Output: There is sufficient statistical evidence to support the company's claim.

Chain-of-Thought Verification:

- Null Hypothesis (H_0): The average response time is ≥ 2 seconds.
- Null Hypothesis (H_0): The average response time is ≥ 2 seconds.
- Alternative Hypothesis (H_a): The average response time is < 2 seconds.
- Alternative Hypothesis (H_a): The average response time is < 2 seconds.

- Validation: The p-value (0.004) is strictly less than the significance level α (0.01).
- Validation: The p-value (0.004) is strictly less than the significance level α (0.01).
- Decision logic: Because $p \leq \alpha$, we reject the null hypothesis.
- Decision logic: Because $p \leq \alpha$, we reject the null hypothesis.

Delta Explanation: The extremely low probability (0.4%) of observing this response time under the null hypothesis validates the shift to the alternative state, verifying the application's high-performance claim.

Analysis 2: Digital Marketing Click-Through Rate:

Scenario Context: A digital marketing team tests an ad to see if it increases the baseline 6% click-through rate (CTR). The resulting p-value is 0.12 at a 5% significance level ($\alpha = 0.05$).

80/20 Core Output: The team should conclude that there is insufficient evidence to claim the new advertisement increases the CTR.

Chain-of-Thought Verification:

- Null Hypothesis (H_0): The CTR is $\leq 6\%$.
- Null Hypothesis (H_0): The CTR is $\leq 6\%$.
- Alternative Hypothesis (H_a): The CTR is $> 6\%$.
- Alternative Hypothesis (H_a): The CTR is $> 6\%$.
- Validation: The p-value (0.12) is greater than the 0.05 threshold.
- Validation: The p-value (0.12) is greater than the 0.05 threshold.
- Decision logic: Because $p > \alpha$, we fail to reject the null hypothesis.
- Decision logic: Because $p > \alpha$, we fail to reject the null hypothesis.

Delta Explanation: Analyzing the performance gap reveals that a p-value of 0.12 implies a 12% probability that the observed CTR fluctuation is due to random noise. The input changes do not conclusively yield a statistically superior output.

Analysis 3: Machine-Learning Model Accuracy:

Scenario Context: An old model has 82% accuracy, and a new model achieves 85%. The statistical test gives a p-value of 0.01.

80/20 Core Output: The new model performs significantly better and should replace the legacy system.

Chain-of-Thought Verification:

- Interpretation of the p-value: A p-value of 0.01 means there is exactly a 1% probability of seeing an accuracy jump of this magnitude (82% to 85%) if the new model was technically identical in predictive capability to the old one.
- Interpretation of the p-value: A p-value of 0.01 means there is exactly a 1% probability of seeing an accuracy jump of this magnitude (82% to 85%) if the new model was technically identical in predictive capability to the old one.
- Decision logic: Since 1% (0.01) is lower than standard acceptability thresholds (like 0.05), the team must reject the null hypothesis.
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Delta Explanation: By utilizing the black box approach to assess structural changes in predictions vs. ground truth, we confidently identify a positive delta in model performance, justifying immediate structural adoption.

Analysis 4: Factory Defect Rate:

Scenario Context: A factory claims a 2% defect rate. Inspection yields a 95% Confidence Interval (CI) of (2.4%, 3.6%) and a p-value of 0.01.

80/20 Core Output: The company's claim of a 2% defect rate must be unequivocally rejected.

Chain-of-Thought Verification:

- CI Logic: We are 95% confident that the true population defect rate is caught between 2.4% and 3.6%. The target of 2% falls entirely outside this window.
- CI Logic: We are 95% confident that the true population defect rate is caught between 2.4% and 3.6%. The target of 2% falls entirely outside this window.
- p-value Logic: The p-value of 0.01 signifies a mere 1% chance of capturing this sample data if the true defect rate were indeed 2%.
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Delta Explanation: Both diagnostic techniques recursively confirm the same truth: the operational baseline has shifted negatively. The true defect rate is significantly higher than the 2% benchmark.

Analysis 5: E-commerce User-Interface Design:

Scenario Context: An online shopping UI is updated. The 95% CI for the difference in average spending is (₹120, ₹480) with a p-value of 0.002.

80/20 Core Output: The new UI design triggers a substantial and statistically significant increase in customer spending.

Chain-of-Thought Verification:

- CI Logic: We are 95% confident that the average spend per customer increases by at least ₹120 (and up to ₹480). Because the entire interval is positive, it confirms a profit increase.

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- p-value Logic: A p-value of 0.002 indicates a 0.2% probability that this spending spike was coincidental.

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Delta Explanation: The interface modifications generated a highly positive delta in consumer transaction behavior. Business Recommendation: Proceed with a full deployment of the new user interface to maximize platform revenue, as the feedback metrics strongly confirm its economic superiority.